RESTORATION ECOLOGY
The Journal of the Society for Ecological Restoration

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RESEARCH ARTICLE

Restoring with forest monocultures: a mechanistic look at invasive species suppression

Clarice M. Esch[1,2], Stephanie G. Yelenik[3]

To create conditions suitable for the recruitment of native forest species, tools are often needed to suppress grasses when restoring forests from abandoned pastureland. Planting trees is one method to reduce grass growth through shading and accumulation of leaf litter, while benefiting woody seedlings by moderating the microclimate. Nitrogen-fixing trees are often planted in tropical reforestation work because of their fast growth rates and ability to enrich soil nutrients. However, nitrogen (N) contributed to the soil from N-fixing trees can also facilitate the growth of competitive invasive grasses. The balance between grass suppression versus facilitation may depend upon tree density and site conditions. Using planted stands of a native N-fixer, *Acacia koa* (koa), on Hawai'i Island, we evaluated the efficacy of invasive grass suppression by koa and what mechanisms might be responsible. We identified a range of tree densities and sites spanning the island. We found consistent effects of grass suppression by koa via shading and litter accumulation, though total grass suppression rarely occurred. There was a positive relationship between soil N and koa basal area, but this did not lead to more grass. It is unclear if the conditions created by dense koa thickets facilitate diverse forest regeneration. If grass suppression to levels that allow for diverse natural regeneration is the desired management goal, then further management interventions could amplify the mechanisms of grass suppression we identified by introducing species that cast deeper shade and/or change the litter environment.

Key words: ***Acacia koa*, Hawai'i, litter accumulation, reforestation, shade, soil nitrogen, succession**

Implications for Practice

- Koa, a tree commonly used in Hawaiian forest restoration, suppresses grass biomass via shading and leaf litter accumulation.
- Despite biomass reductions, invasive grasses persist at high cover levels under koa canopies, and these levels may stall further forest succession. If grass suppression and/or more diverse forests are management goals, then additional interventions are likely needed.
- Mechanisms of shading and leaf litter accumulation could be reinforced by densely planting additional native woody species that are not nitrogen fixing.

Introduction

Few plant communities in the world are free from invasive species, making management of plant invasions a universal challenge for many restoration projects, depending upon restoration goals (D'Antonio et al. 2016). Invasive plants can displace native species and thereby alter community composition and ecosystem functioning (Mack et al. 2000; Levine et al. 2003). Invasive pasture grasses in particular are frequently introduced by humans for grazing livestock and have become widespread and invasive over large areas (D'Antonio & Vitousek 1992). These grasses can be a substantial barrier to forest restoration by competing with desired woody species, inhibiting native recruitment, altering fire regimes, and exhibiting priority effects that lead to continued grass dominance after pasture abandonment (D'Antonio & Vitousek 1992; Holl et al. 2000). Therefore, methods that can be used over large scales are needed to effectively control invasive pasture grasses during restoration efforts.

One method is planting native trees, which can work to suppress grasses through multiple mechanisms (Parrotta 1992; Holl et al. 2000). Shade cast by trees can reduce grass growth because grasses are primarily shade-intolerant (Holl et al. 2000). Physical accumulation of tree leaf litter can negatively affect herbaceous vegetation, including grasses (Xiong & Nilsson 1999). As trees mature, their canopies moderate daily temperature and moisture extremes and create less stressful environments, which can benefit the recruitment of additional woody species (Holl 1999; Rehm et al. 2021).

Nitrogen-fixing (hereafter N-fixing) trees are common candidates for tree planting in many different tropical regions because they grow quickly and rapidly form canopy structure (Scowcroft et al. 2004; Griscom & Ashton 2011). They can benefit plant

Author contributions: CME, SGY conceived and designed the research; CME performed the experiments and analyzed the data; CME, SGY wrote and edited the manuscript.

[1]USDA Forest Service, Shawnee National Forest, Vienna, IL, U.S.A.
[2]Address correspondence to C. Esch, email clarice.esch@usda.gov
[3]USDA Forest Service, Rocky Mountain Research Station, Reno, NV, U.S.A.

doi: 10.1111/rec.70162
Supporting information at:
http://onlinelibrary.wiley.com/doi/10.1111/rec.70162/suppinfo

growth by contributing soil organic matter and N, primarily through decomposition of their N-rich leaf litter, particularly in sites that have been degraded due to overgrazing or other agricultural practices (Scowcroft et al. 2004; Batterman et al. 2013). Soil N contributions can be crucial in overcoming N-limitation, which occurs in many younger tropical soils (Vitousek & Farrington 1997; Batterman et al. 2013). Grass suppression via tree shading and litter deposition, plus improved conditions for tree seedling recruitment (moderated microclimate, increased soil N and organic matter) may create conditions for passive restoration of forest communities following initial tree planting (Corbin & Holl 2012; Yelenik et al. 2021).

However, planting N-fixing trees may not always successfully suppress pasture grasses and lead to forest succession, especially since many mechanisms that enhance native species growth can also facilitate invasive species (Gilbert & Lechowicz 2005; Yelenik et al. 2022a). Increased soil N driven by N-fixers can benefit grasses to the detriment of native species (Maron & Connors 1996; Sierra & Nygren 2006). Decomposition rates impact the thickness of the litter layer and associated grass suppression; the N-rich leaf litter of N-fixing trees decomposes rapidly, creating thinner litter layers less likely to suppress grasses than the litter of co-occurring non-N-fixing species (Scowcroft 1997; Tateno et al. 2007; Yelenik et al. 2022b). Decomposition rates and ecosystem N cycling rates may also be increased by higher foliar N in grasses from N-fixing tree inputs (Perakis et al. 2012). Thus, a mixture of mechanisms simultaneously suppressing and facilitating grasses could occur under N-fixing trees.

The balance between grass suppression and facilitation by trees may shift depending upon tree density or basal area (Callaway & Walker 1997). Increasing basal area should intensify grass suppression via shading and litterfall, but increased N inputs could also facilitate grasses and counteract suppressive effects from higher basal area (Sitters et al. 2013). Furthermore, as basal area and associated shading increase in transitions from savanna to forest, grass composition shifts to more shade-tolerant species (Charles-Dominique et al. 2018). Thus, when reforesting pasture, grass suppression by tree canopies could be undermined as shade-intolerant grass species are replaced by more shade-tolerant grasses (McDaniel & Ostertag 2010).

To examine interactions between a N-fixing tree and invasive grasses in restoration, we studied the native Hawaiian leguminous species koa (*Acacia koa* A. Gray). Hawaiian forests are relatively low in tree diversity, with koa dominating the canopy in more than 10% of forest on Hawai'i Island (Baker et al. 2009). Koa grows quickly and is easy to propagate relative to other native Hawaiian tree species, making it common to plant in monoculture for reforestation in Hawai'i (Jeffrey & Horiuchi 2003; Nelson et al. 2024). Goals of such reforestation efforts in old pasture systems often include natural secondary succession to a more diverse forest. However, invasive grasses often persist in the understory of koa plantings, impeding the recruitment of native forest species, and thereby stalling succession in a koa–grass state (Denslow et al. 2006; McDaniel et al. 2011; Yelenik et al. 2021). Disturbances such as fire often reverse any biodiversity gains and maintain the koa–grass state, highlighting a challenge of restoring this system (Hamilton et al. 2021).

Across the Hawaiian landscape, steep topography contributes to widely variable climatic conditions across relatively small expanses (Giambelluca et al. 2013), which may lead to disparate restoration outcomes when koa is used to suppress grasses. Variation in climate could add further complexity to the balance between competition and facilitation. For example, rainfall and temperature can affect decomposition rates (Austin 2002; Powers et al. 2009), which in turn alters the thickness of the litter layer and the rate at which N is contributed to the soil. The outcomes of tree–grass interactions may be dependent upon how individual mechanisms are influenced by gradients in tree basal area and/or climate, which makes understanding these dynamics critical for successful forest restoration.

To examine the efficacy of grass suppression by koa monocultures, we asked the following questions: How does grass biomass change with koa basal area? and what mechanisms (e.g. decomposition, climate variables) appear to be responsible for the relationship between koa and grass biomass? We sampled grass biomass under and away from koa canopies at sites undergoing forest restoration that span a range of temperature and rainfall across Hawai'i Island (Table 1). Widely variable topographic and climatic gradients across the island make it difficult to select sites that alter a single component in isolation, but we expect average annual rainfall and temperature to impact koa–grass interactions. Relating to *climate variables* (rainfall and temperature) we hypothesized that: (a) warmer and/or wetter sites would have higher grass production and koa basal area; (b) faster decomposition in hotter or wetter sites may reduce the accumulation of litter. Relating to *koa–grass interactions* across sites, we hypothesized that (c) grass biomass would decrease with increasing koa cover because higher basal area of koa should lead to less light availability, lower soil moisture, and greater litterfall leading to a build-up of koa litter in the understory. Alternatively, (d) greater koa basal area may lead to increasing soil N availability, which may facilitate greater grass growth and faster litter decomposition, and counteract competitive effects if grasses are shade-tolerant.

Methods

Study System

We studied koa–grass interactions in koa-dominated secondary forest stands on Hawai'i Island, USA. In Hawai'i, like many other tropical and subtropical areas, much of the forest was cleared and converted to pasture for grazing livestock in the 1800s and into the 1900s (McDaniel & Ostertag 2010; Corbin & Holl 2012). Common invasive grasses on Hawai'i Island and our study sites are meadow ricegrass (*Ehrharta stipoides* [Labill.] R. Br.) and kikuyu grass (*Cenchrus clandestinus* [Hochst. ex Chiov.] Morrone) (McDaniel & Ostertag 2010; Yelenik 2017). In recent decades, ranching has become less profitable, leading to the abandonment of pastureland and a growing interest in reforestation to increase habitat for endangered birds and aid other ecosystem services (Scowcroft & Jeffrey 1999). Koa is a canopy dominant or codominant tree occurring on many of the islands and spanning a range of elevations, soil types, and rainfall regimes

Table 1. Characteristics of the 10 study sites of koa-dominated restoration stands ranging across Hawai'i Island.

Site	*Approx. Elevation (m)*	*Avg. Annual Temp. (°C) (Giambelluca et al. 2014)*	*Avg. Annual Rainfall (mm) (Giambelluca et al. 2013)*	*Koa Basal Area (m^2/ha) Range*	*Restoration History*
Hakalau	1845	12.1	2,559.7	6.5–44.7	Koa was planted (1985–1990) in corridors extending from intact forest within fencing (Jeffrey & Horiuchi 2003)
Kahuku 1	1,105	16.5	1,358.6	2.2–126.1	Fenced in 2005 with various restoration treatments (herbicide, soil turnover, planting, and seedling) applied (McDaniel et al. 2011)
Kahuku 2	1,292	15.4	1,302.5	0.4–162.3	
Kahuku 3	1,357	14.9	1,522.8	5.3–116.1	
Hilina Pali 1	732	19.4	1796.6	0.8–88.4	Regenerated from 1950s koa plantings after fires in 1970s and 1980s (Tim Tunison, National Park Service, pers. comm. 2013)
Hilina Pali 2	846	18.3	2037.3	2.2–67.2	
Kona Hema 1	1728	12.3	877.3	1.9–42.6	Fenced in 2004, koa was planted and regenerated from existing trees (Giffin 2017)
Kona Hema 2	1,456	14.2	808.8	2.7–56.7	
Pu'u Wa'awa'a 1	1,319	15.5	702.74	0.8–52.0	Fenced in 1985, koa regenerated from existing trees (Giffin 2010)
Pu'u Wa'awa'a 2	1,334	14.7	709.08	0.3–66.9	

(Wagner et al. 1990). Koa is highly valued, both culturally and commercially (Elevitch et al. 2006), and grows relatively quickly, leading to its widespread use in reforestation. In addition, koa can resprout from root suckers, making it a species that often becomes dominant under natural succession after grazer removal, particularly if remnant koa are present (McDaniel et al. 2011; Scowcroft & Yeh 2013).

Site Descriptions

We established 10 study sites across five protected areas encompassing a wide range of climatic conditions on Hawai'i Island (Table 1; Fig. 1). These protected areas are managed by a diverse group of landowners and stakeholders, showing the ubiquity of koa as a restoration tool across Federal, State, and Non-Profit agencies. Some sites are within the same protected area but are spatially separated, and owing to steep climatic gradients, experience distinct temperature and moisture regimes. Though the protected areas have varying restoration and land use histories (Table 1), the sites are similar in that they contain koa-dominated stands on land occupied by invasive grasses disturbed by some combination of deforestation, burning, grazing, and/or logging. Most of the sites have similar soils and substrate ages (750–3,000 years old), except Hakalau, which is significantly older (>10,000 years; Table S1). To exclude feral ungulates, all sites except Hilina Pali 1 and 2 are fenced. However, maintaining intact fences is difficult, and ungulates, particularly pigs, likely have some influence at all sites. When selecting plots, we avoided any areas with recent damage by ungulates.

Data Collection

Plot Selection. At each site, we established fifteen 2.5-m radius circular plots (each plot approximately 20 m^2), with at least 20 cm soil depth, and located them to maximize the range

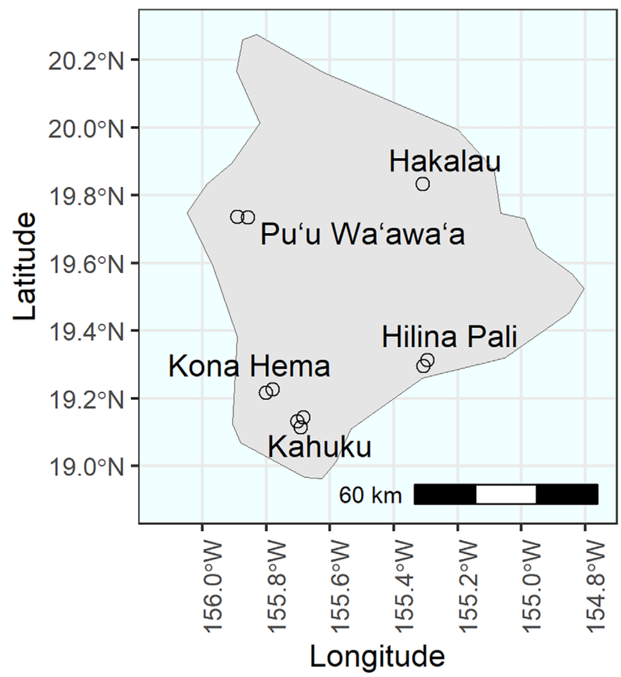

Figure 1. Map of site locations on Hawai'i Island.

of koa densities sampled at each site (Table S2; Fig. S1). Plot edges were separated by at least 10 m, except at Kahuku 3 where the higher-density plots were separated by at least 5 m because few areas of high-density koa existed. Of the 15 plots at each site, 10 plots included koa stems and grass (hereafter koa plots) and 5 control plots contained only grass and no koa stems or canopy cover. Our control plots thus did not experience light reduction or resource uptake by koa, but it is possible they experienced koa litterfall from nearby trees, which may increase N cycling relative to pasture with no nearby koa (D'Antonio et al. 2024). Woody species rarely occurred in the understory of these sites unless planted (pers. obs.); to avoid any confounding effects of tree species other than koa, all plots contained only herbaceous species in the understory.

To stratify sampling across a given site, plots were selected in clusters of three: two koa plots (one high density and one low density) and one control plot. Control plots were in the vicinity of koa plots in open spaces in the stand. The exception to the plot clustering method was Kahuku (1, 2, and 3); at these sites we distributed plots across the limited space (200 m × 200 m) of the restored area. High- versus low-density koa was relative within a site and plot cluster, and we aimed to capture a wide range of possible stem densities (1–79 stems/plot) and basal area (0.3–162.3 m^2/ha). Thus, density range varied by site, for example, at dry sites koa grows less densely and has a more open growth habit leading to a lower maximum plot stem density than at wet sites. We quantified the effect of koa on grasses using both stem density and basal area to incorporate the effect of tree size. We measured diameter at breast height (DBH) for all koa stems exceeding 0.5 cm at breast height (1.37 m) in each plot. These DBH values were then used to determine the basal area of koa per plot.

The following measures were made in each plot at every site.

Light (Photosynthetically Active Radiation). We measured photosynthetically active radiation using a hand-held light wand (MQ-301, Apogee Instruments, Inc., Logan, UT) on clear to minimally overcast days to assess light availability near the ground. Clear to minimally overcast days are the conditions most commonly encountered at our sites during the dry season; thus, these are the conditions under which we took our measurements, except at Pu'u Wa'awa'a, where a cloudless sky was not encountered and light measures were taken on a uniformly overcast day. At all sites, we averaged five measures per plot—one at plot center and then at four points 1 m from plot center. The light wand was held at ~1 m height, exceeding grass height. Measures were made when the sun was unobscured by clouds and directly or nearly directly overhead to capture shade cast only by the plot's vegetation.

We assessed light under koa canopies as a percent of light available in the open at approximately the same time. Thus, to calculate the percent of full sun available we divided the average amount of light in a koa plot by light levels in absence of canopy (usually the nearest control plot). All measures took place within a month (June 24, 2019 to July 24, 2019). Light levels were slightly greater in four koa plots than in their nearest control plot, likely due to some small clouds passing overhead when measuring the control plot; for these koa plots light levels were truncated at 100% full sun.

Soil Mineral Nitrogen. To measure the influence of koa trees on soil N, we sampled soil from 0 to 10 cm depth using a spade from three haphazardly selected points at least 0.5 m apart from each other within each plot. Samples from each plot were combined, homogenized, refrigerated, passed through a 4.75-mm mesh sieve, and processed within 48 hours of collection. To extract nitrate and ammonium, subsamples of field wet soil were shaken with 2M KCl for 1 hour. Extracts were analyzed colorimetrically (ELx808 Absorbance Microplate Reader, BioTek Instruments, Inc., Winooski, VT) (Sinsabaugh et al. 2000; Doane & Horwáth 2003). We measured inorganic soil N again following an approximately 5-week incubation to calculate N mineralization. We prepared the incubations by placing 50 g of each soil sample in loosely capped jars and adding deionized water to all samples to homogenize soil moisture levels to the wettest sample (3.2 g H_2O/g dry soil). Incubating samples were kept in a cooler in the Magma Lab at Hawai'i Volcanoes National Park (average temp. = 16°C) to modulate temperatures.

Soil Moisture. To measure soil moisture in the field, we used a soil moisture probe (Hydrosense II, Campbell Scientific, Inc., Logan, UT) inserted at three points to 12 cm depth within each plot. The probe was placed at least 0.5 m from the bole of any tree over 10 cm DBH to avoid roots and inserted into haphazardly placed locations (at least 0.5 m apart from each other) with at least 12 cm of soil. We avoided taking measurements within 2–3 days of rainfall events. We took these measures three times over the course of the study, roughly 1.5 months apart to capture variation in soil moisture while transitioning into the dry season, and then averaged measures for analysis.

Decomposition. To examine decomposition rates and determine the speed at which the litter layer turns over, we used a tethered leaf method (Vitousek et al. 1994). Mature koa trees rarely produce true leaves and typically have canopies composed of phyllodes (sickle-shaped modified petioles) (Baker et al. 2009). When collecting koa leaves for measures of decomposition, we used only phyllodes, and for simplicity, we use the term koa leaves to refer to phyllodes. We collected fully expanded koa leaves grown in full sun at each site and placed them in a drying oven at 35°C for at least a week to dry them to a consistent state without brittleness. We weighed each leaf individually before tethering them together in strands of five leaves. Strands of tethered leaves were buried underneath leaf litter and grass rhizomes, so contact was made between our leaves and the soil. At each plot, we placed four strands of five leaves strung together with a washer at each end. All strands were centered on the same point, and individual strands were spread apart to avoid touching each other. One string of leaves was removed after each consecutive month for a total of 4 months. After retrieval, the leaves were dried at 60°C, weighed individually, and subtracted from their initial mass (converted to dry mass) to calculate change in mass. A dry mass conversion from starting mass was established by drying additional leaves from each site at 35°C and then 60°C. The percent change in mass for all leaves on a strand was then averaged and used to determine the rate of decomposition per plot using the equation: $\text{mass}_t = \text{mass}_0 \times e^{-kt}$, where k is the litter decay constant, t is time, and 0 is time point zero (starting time) (i.e. $mass_0$ is the starting mass of the leaf), and e is a constant.

In the first month, a few leaves (eight) gained mass, which may be due to variation in initial drying. We converted the mass gain of these leaves to 0.01% mass loss. In the third and fourth months, some leaves became untethered, or substantial parts broke off and could not be retrieved. Often, other leaves were retained on a strand, and decomposition data from that plot could still be collected, just with fewer leaf replicates on a strand. Six whole strands of leaves (out of 480) were lost, resulting in missing data for a plot for a month.

The decomposition study was pursued at all sites except at Pu'u Wa'awa'a 1 and 2 because these sites were added after the 4-month decomposition study began.

Grass and Koa Litter Biomass. We measured biomass by removing all plant material within a 0.6-m × 0.6-m quadrat in each plot and drying it at 70°C for at least 48 hours. The quadrat was placed haphazardly such that it was representative of plot vegetation. We sorted the dried samples into grass, koa leaf litter, 'ōhi'a (*Metrosideros polymorpha* Gaudich.) leaf litter, coarse woody debris, and other material (usually seed pods and flowers) and weighed each component.

Percent Cover/Species Composition. The lead author visually estimated percent cover of herbaceous vegetation and woody species litter in 0.6-m × 0.6-m quadrats. The quadrat was haphazardly placed three times in each plot to be representative of the vegetation. We identified all species of grass and herbaceous plants and assessed percent cover for each species or litter type.

Leaf C and N Content. At each site, we collected fully expanded koa leaves grown in full sun to assess % C and N content. We avoided leaves damaged by herbivores or already turning brown. Leaf samples were processed at the University of Hawai'i Hilo Analytical Laboratory using an elemental analyzer (Costech Analytical Technologies Inc., Valencia, CA).

Statistical Analysis. We completed all analyses using R version 4.2.0 (R Core Team 2022). To account for the high variability of koa stem density, within and across sites, we used basal area as a continuous variable for analysis.

To address Hypotheses a and b, we developed generalized linear models with a gamma distribution. For Hypothesis a, we used site-level means of grass biomass or koa basal area as response variables, with climate variables as predictors. For measures of grass biomass, only control plots were used. For measures of koa basal area, only koa plots were used. For Hypothesis b, we used site-level means of decomposition rate or koa litter biomass as response variables, with climate variables as predictors. We also modeled koa litter biomass as a response to decomposition rate, using plot-level measures, with site as a random effect.

To evaluate the effects of koa basal area on grass biomass and the mechanisms involved (outside of climate variables), we developed a Structural Equation Model (SEM) using the piecewise SEM package (version 2.1.0) (Lefcheck 2016). Components of this analysis address both Hypotheses c and d. SEMs are useful for understanding direct and indirect pathways (Grace et al. 2012), making them ideal for our system due to the many ways our factors of interest could influence each other and grass biomass. Our initial SEM was based on knowledge of the system and hypothesized relationships between koa and grass, including processes like decomposition (Table S3; Fig. 2A) from 150 plots (100 koa plots and 50 control plots) across 10 sites. Each component model within the SEM used a gamma distribution and included a random effect for site. Two plots at Kahuku 3 did not include any grass and were modified to include 0.0001 g grass biomass in order to be modeled with a gamma distribution, which does not include 0.

We evaluated indirect paths between koa basal area and grass biomass mediated by light, litter, soil N (or N mineralization rate), litter decomposition rates, and soil moisture. We selected the SEM that best fit our data by examining model Akaike Information Criterion and Fisher's C to determine global goodness of fit. We added a single path from koa basal area to grass biomass,

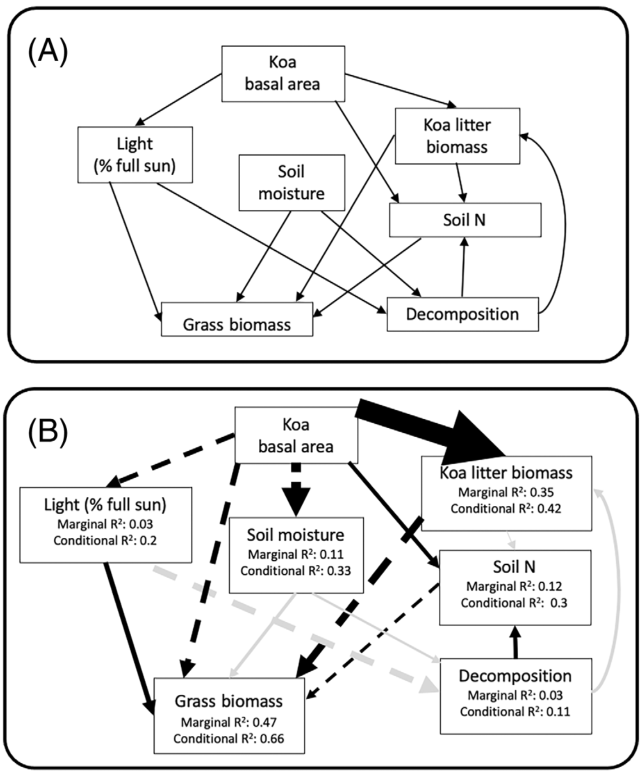

Figure 2. Initial (A) and final (B) SEM evaluating the effects of koa basal area on grass biomass and possible mechanisms. Black arrows signify statistically significant ($p < 0.05$) paths and gray arrows nonsignificant paths. Solid arrows denote positive relationships and dashed arrows denote negative relationships. Arrow width scales with standardized path coefficients. Marginal R^2 includes variance from fixed effects, and conditional R^2 considers both fixed and random effects.

which improved model fit based on tests of directed separation (Lefcheck 2016). Path coefficients were standardized by scaling the raw data prior to analysis. Data were scaled to have a standard deviation of 1 and centered on 2 to avoid negative values that would have resulted from 0-centering, as negative values cannot be modeled with a gamma distribution.

To evaluate shifts in understory cover, including percent cover of litter and grass (and address components of Hypotheses c and d as to whether koa facilitates or competes with grasses), we used a PERMANOVA with koa basal area and light as fixed effects and site as a random effect using the adonis function in R's vegan package (Oksanen et al. 2019).

Results

Effect of Climate on Grass Production and Koa Basal Area

We expected (Hypothesis a) grass production and koa basal area might shift in predictable ways across climatic gradients, but we did not find consistent patterns (Fig. S2). For grass production in the absence of koa canopy cover (control plots), there were no strong relationships with temperature or rainfall. Although we found that annual average temperature was significantly related to grass biomass ($p = 0.02$; Table S4a), this was driven by Hilina Pali 1 and 2 (Table S4b). These sites are the warmest and only sites with molasses grass, *Melinis minutiflora* P. Beauv, which grows substantially larger than the grass species present at other sites. Koa basal area was not related to temperature or rainfall (Table S4c).

Effect of Climate on Decomposition and Koa Litter

We expected (Hypothesis b) decomposition rate and koa litter biomass might be related to climatic gradients, but found no relationships (Fig. S3). Decomposition rates were consistently variable across sites, regardless of temperature or rainfall (Table S5a). Similarly, koa litter biomass was not consistently related to climate variables (Table S5b). Koa litter biomass, however, was positively correlated with decomposition rates (Table S6; Fig. S4).

Given the lack of relationships between climate variables and grass biomass, koa basal area, decomposition, and koa leaf litter, climatic gradients in and of themselves do not seem to have consistent effects on koa–grass interactions across our sites.

Mechanisms of Koa–Grass Interactions

We used a SEM to look at direct and indirect effects of koa basal area on grass biomass and the mechanisms that help shape these relationships. Generally, grass biomass decreased with increasing koa basal area (Hypothesis c) (Figs. 2B & 3). As hypothesized, both decreased light and greater koa litter biomass were associated with grass suppression from increasing koa basal area (Table S7; Fig. 2B). Koa litter biomass increased with koa basal area ($r = 0.322$, $p < 0.0001$) and additional litter led to diminished grass biomass ($r = -0.123$, $p < 0.0001$) (Fig. 4A). Light reaching the understory decreased as koa basal area increased ($r = -0.081$, $p = 0.02$) and decreasing light reduced grass biomass ($r = 0.083$, $p = 0.0025$) (Fig. 4C). Although koa basal area had a negative effect on soil moisture ($r = -0.158$, $p < 0.0001$) consistent with Hypothesis c, soil moisture had no subsequent effect on grass biomass ($r = 0.056$, $p = 0.26$) (Fig. 4B).

The direct path from koa basal area to grass biomass was an addition to improve fit. This added path indicated there are further negative effects of koa on grass biomass ($r = -0.094$, $p < 0.0001$) not mediated through measured factors in the model.

Differences between sites accounted for a substantial portion of variation in the SEM, as shown by the increase from marginal R^2 (accounting for fixed effects alone) to conditional R^2 values (includes site random effect and fixed effects) (Fig. 2B). Conditional R^2 values were 0.07–0.22 units greater than marginal R^2 values. Variation between sites in the effect of increasing koa basal area on grass biomass can be easily seen in Figure 3. Generally, increasing koa basal area had a negative effect on grass biomass, though there was often at least one plot at each site where grass biomass increased in the presence of koa relative to nearby controls without koa canopy cover (Fig. 3). Kahuku 1, 2, and 3 were the exceptions, where grass biomass decreased at all koa plots (Fig. 3). Kahuku 3 contained the two plots (out of 100 koa plots) where grass was entirely eliminated. The Kahuku sites also reached the highest basal area of any of our plots (Table 1).

Consistent with Hypothesis d, soil N was higher with increasing koa basal area ($r = 0.066$, $p = 0.02$). However, inorganic soil N had a negative relationship with grass biomass ($r = -0.06$, $p = 0.0348$), which is opposite the expected relationship. Also, the low r values show that little variability was explained by these relationships. Also counter to our expectations, decomposition rate did not affect koa litter biomass ($r = 0.059$, $p = 0.1$) and was not influenced by light ($r = -0.1$, $p = 0.1$) or soil moisture ($r = 0.045$, $p = 0.57$). Decomposition rate was also not related to leaf C:N ratio ($r = -0.012$, $p = 0.9$; Table S8; Fig. S5). We also ran the same SEM using N mineralization rate. We hypothesized that soil N availability may be associated with greater grass biomass (Hypothesis d), but this was not the case ($r = -0.020$, $p = 0.52$). N mineralization was also not influenced by koa basal area or koa litter biomass; though it was related to decomposition rate ($r = 0.075$, $p = 0.04$).

Grass cover and composition as well as koa litter cover shifted with koa basal area and the amount of light reaching the understory (Table S9; Fig. S6). Specifically, koa litter cover increased and grass cover decreased with koa basal area. Cover of *Cenchrus clandestinus*, a relatively shade-intolerant species (McDaniel & Ostertag 2010), decreased under koa canopies (relative to control plots) at the eight sites where it was present, whereas *E. stipoides* (a shade-tolerant species; McDaniel & Ostertag 2010) cover increased under koa canopies at five of the six sites where it was present (Table S10).

Discussion

Our results demonstrate that increasing koa basal area is associated with decreased invasive grass biomass within and across

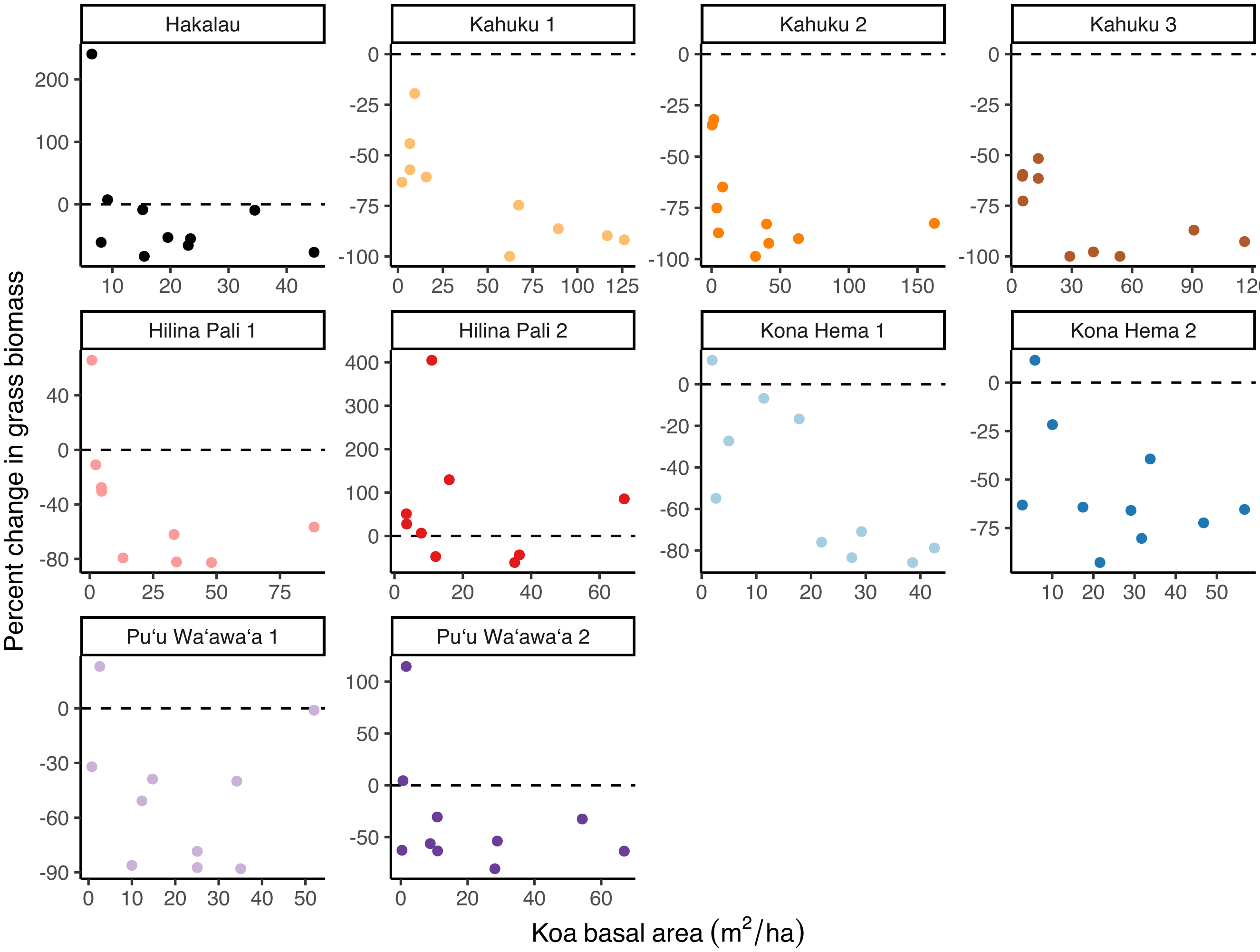

Figure 3. Relationship between koa basal area and percent change in grass biomass relative to control (no koa canopy cover) plots at each site.

sites varying in temperature and rainfall. While we found that soil N increased with koa basal area, counter to expectations it was not associated with greater grass biomass. Rather, light reduction (presumably due to great canopy cover) and greater koa litter deposition, both associated with high koa basal area, were found to be more important than changes in soil N in suppressing invasive grasses. Although the presence of koa generally suppressed grass and reductions in grass biomass were significant, invasive grass cover often remained, even under relatively dense koa canopies. A shift in composition to more shade-tolerant grass species under dense koa canopies might allow grasses to maintain dominance (McDaniel & Ostertag 2010). This in turn perpetuates koa–grass states because invasive grasses reduce recruitment of native forest species (Denslow et al. 2006; Rehm et al. 2023; Yelenik et al. 2024). Our sites ranged across Hawai‘i Island, encompassing varying abiotic conditions that could affect the efficacy of grass suppression by koa. The lack of a relationship between grass suppression and climate variables could be due to the relatively short-term nature of this study (<6 months) and interannual variability in climate. Given the breadth of variation our sites span in terms of climate and grass species, our findings are relatively applicable across the island and show consistency in the mechanisms involved in grass suppression. More generally, our results provide evidence that a stable state (koa–grass forests, in this case) is held in place through a complex set of interactions. Therefore, finding a single mechanism that might help push the system into a different basin of attractions may not be a simple process (Suding et al. 2004; Fukami & Nakajima 2011).

Mechanisms of Grass Suppression

Multiple mechanisms contribute to grass suppression by koa. Increasing koa basal area decreased light levels, which in turn reduced grass biomass. This finding aligns with observations of reduced grass biomass with decreasing light levels on Hawai’i Island (McDaniel & Ostertag 2010; D’Antonio et al. 2024) and the importance of canopy cover in diminishing competition with pasture grasses in reforestation work

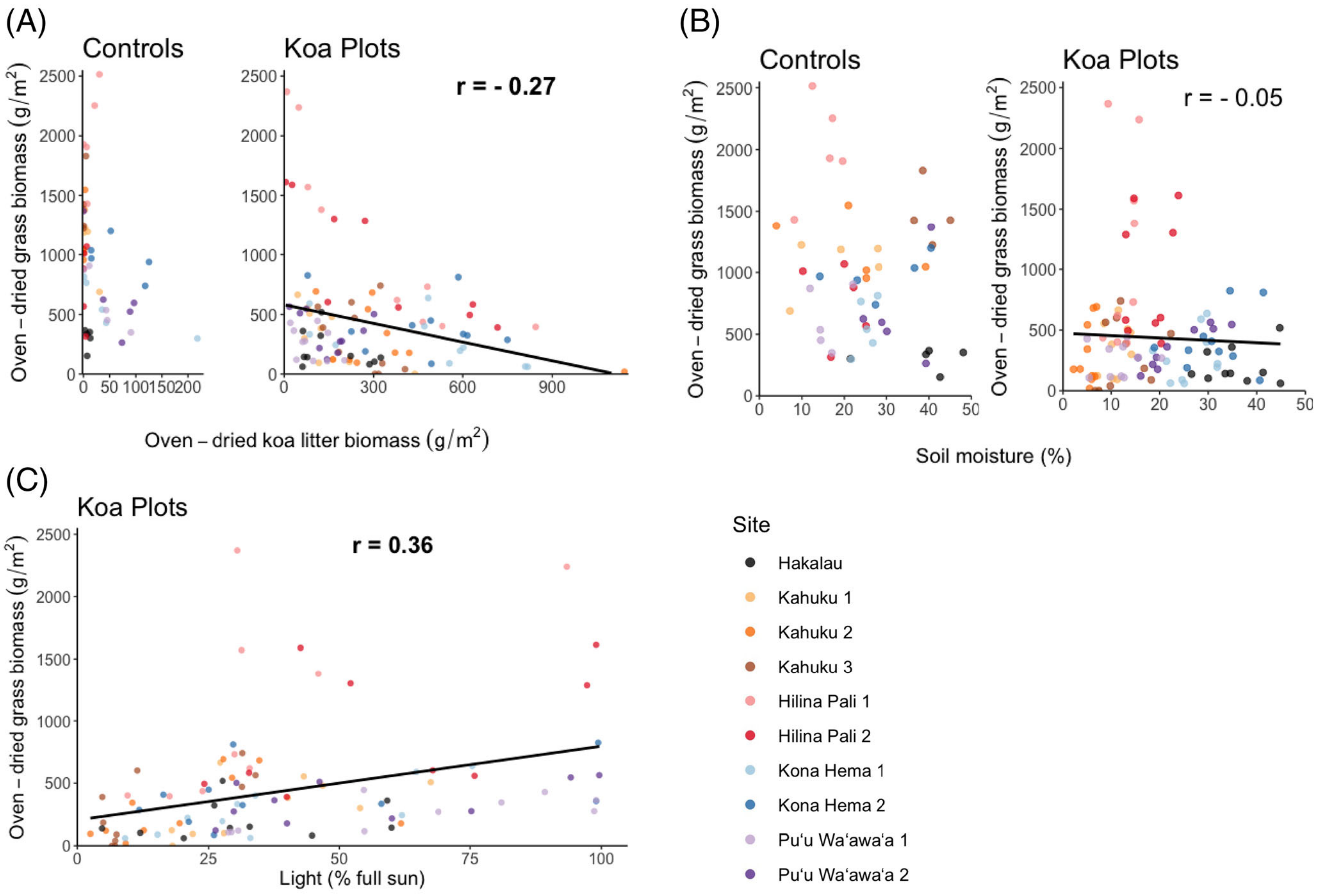

Figure 4. Bivariate correlations between grass biomass and (A) koa litter, (B) soil moisture, and (C) light in control (only grass) and koa plots. Bolded *r* values are significant Pearson's correlations ($p < 0.05$).

throughout the tropics (Holl et al. 2000). Light reduction to 5% full sun, which substantially reduces grass growth in this system (McDaniel & Ostertag 2010), was infrequently reached at our sites. Planted koa stands typically reduce light to 10–20% full sun (Scowcroft & Jeffrey 1999), which is relatively high light compared to approximately 1.9–10% full sun in the understory of intact Hawaiian forests (Burton & Mueller-Dombois 1984) and other tropical forests globally (Record et al. 2016). Relatively high light transmission through koa canopies may be driven by the orientation of koa leaves. Most mature koa trees have canopies filled with phyllodes, not true leaves, which have chlorophyll on both sides and tend to be oriented vertically to capture light and conserve water (Baker et al. 2009). It is also important to note, while koa canopies reduce light levels and diminish grass biomass, shading is not a mechanism acting in isolation.

As leaves fall from koa canopies and accumulate into a litter layer, this layer also suppresses grasses. In our study, leaf litter accumulates as koa basal area increases, which in turn reduces grass biomass. Light is substantially reduced beneath litter (Vazquez-Yanes et al. 1990). Leaf litter can physically impact understory vegetation by reducing the establishment of new individuals and directly shading existing grass at the forest floor (Amatangelo et al. 2008; Barbier et al. 2008). Similarly, in a temperate forest, leaf litter reduced light reaching the soil and decreased the density of two annual grasses (Facelli & Pickett 1991). Here we found that koa leaves, both when in the canopy and after falling to the ground, were associated with decreased grass biomass.

Unexpectedly, the litter decomposition rate was not related to light or soil moisture. Though substantial koa leaf decomposition can occur in a 4-month span (Yelenik et al. 2022b), the rate of decomposition in the present study was likely limited by the dry season, as decomposition increases with precipitation (Powers et al. 2009). Measures of soil moisture were made with a probe and limited to three points in time, rather than continuous measures from a datalogger, which could also have influenced these results.

Mechanisms of Grass Persistence

Though we found a positive relationship between soil N and koa basal area (consistent with Hypothesis d), higher soil N did not facilitate greater grass biomass. The lack of an effect of N on grasses runs counter to expectations of an association between N-fixing trees and facilitation of understory vegetation (Blaser et al. 2013) and prior work suggesting that high soil N under koa canopies drives persistence of grasses (Yelenik 2017). However, Yelenik (2017) did not quantify light levels and studied solitary koa and 'ōhi'a trees where grasses experience a high light environment relative to that found under the range of koa densities examined in the present study. Other work at Hakalau examining the effects on soil N of litter additions and varying densities of understory plantings found that although dense understory plantings can suppress grass via changes in the light environment, soil nitrogen levels are difficult to alter (Yelenik et al. 2022b; D'Antonio et al. 2024). Phosphorus could also become limiting in the presence of abundant N fixed by koa (Scowcroft et al. 2007, but see Scowcroft et al. 2008) and other *Acacia* spp. (Ludwig et al. 2004; Sitters et al. 2013). Nevertheless, our study shows that suppression via litterfall and shading overwhelms any benefits for grasses of higher soil N with increasing koa basal area (Pearson correlation: $r = 0.23$, $p = 0.02$). Thus, when considering the balance between competition and facilitation in koa–grass interactions (Callaway & Walker 1997), the net effect of koa basal area on invasive grasses across our sites is negative and higher soil N does not appear to aid grasses.

A shift in grass composition to more shade-tolerant species might act counter to mechanisms of grass suppression. We found understory cover composition changed as light and basal area increased. Though we selected plots to exclude any woody species in the understory, herbaceous cover (primarily grasses) varied. The shift in understory cover composition is likely driven in part by increasing koa litter cover but also a transition from shade-intolerant grasses, like *Cenchrus clandestinus*, to more shade-tolerant grasses, such as *Ehrharta stipoides*, just as McDaniel and Ostertag (2010) observed at Hakalau. Thus, the mechanisms we observed might drive suppression of some grasses which are subsequently replaced by other more tolerant species in a pattern of continual or secondary invasion (D'Antonio et al. 2017; Nsikani et al. 2019).

Although planting koa monocultures and increasing koa basal area offer routes towards grass suppression, grass remains present in the understory of nearly all our plots (98 out of 100), much like other areas of regenerating koa (Scowcroft et al. 2008). When seeds of native woody species are present, recruitment can occur once a threshold of ~50% reduction in grass cover is met (Rehm et al. 2023). Depending on the site, many of our koa plots did pass this approximately 50% threshold, and caution should be taken that grass suppression can lead to secondary invasions (S. Yelenik, pers. obs.).

Grass cover is linked to negative effects on seedlings in tropical (Cabin et al. 2002a; Denslow et al. 2006) and temperate forests (Flory & Clay 2010) and little to no passive recruitment of woody species occurs among grasses under koa (Yelenik 2017; Rehm et al. 2019; Yelenik et al. 2021). In addition, it is possible that the high koa basal area also incurs negative effects on incoming native seedlings via competition. Thinning studies in koa stands showed that a combination of stand thinning and grass removal was needed to see positive growth effects on koa itself, suggesting a highly competitive environment for other native woody species as well (Scowcroft et al. 2007). Regardless of ratios of koa and grass in these secondary forests, they seem to be stable, low-diversity ecosystems, as evidenced from other studies (Hamilton et al. 2021; Yelenik et al. 2022b; Rehm et al. 2023).

A suite of management options can reinforce the mechanisms of shading and litter accumulation and/or directly reduce grass biomass, while fostering greater native woody diversity. Shading can be enhanced by incorporating additional understory tree, shrub, and fern species into koa stands, which could create a more structurally complex canopy and might improve the ability of these restoration forests to reduce light levels and grass biomass (Pretzsch 2014). The effect of the litter layer on grass suppression can be strengthened by planting other native Hawaiian tree species, such as 'ōhi'a, that produce litter that decomposes slower than koa litter (Scowcroft 1997; but see Yelenik et al. 2022b), leaving greater standing litter biomass on the forest floor (Yelenik 2017). Planting woody and fern understory species at high densities in forest patches (i.e. the applied nucleation method, sensu Corbin & Holl 2012) has already been found to reduce grass biomass and may also increase litter (D'Antonio et al. 2024) and lead to natural regeneration. In the D'Antonio et al. (2024) study, high-density plantings of diverse species reduced grass biomass by half as compared to low-density plantings, which themselves reduced grass biomass to half if no understory was planted under koa at all. Grass biomass can also be directly reduced through techniques like applying herbicide or plastic mulch (Cabin et al. 2002b; Pinto et al. 2015).

Similar to conclusions from Yelenik (2017), this study suggests that planting or facilitating a diverse array of species in the understory to create a more structurally complex canopy, take up more resources, and increase litter inputs will ultimately help reduce invasive grass biomass (Levine et al. 2004; Chazdon 2008). The benefits of jumpstarting succession by planting a diverse assemblage will likely extend beyond grass suppression and enhance ecosystem function and productivity (Chazdon 2008; Tilman et al. 2014).

We identified two mechanisms by which increasing koa basal area causes the net effect of suppressing grasses: shading and litter accumulation. The mechanisms of shading and litterfall were consistent across sites spanning a range of abiotic conditions and varied site history, demonstrating their ubiquity. Though climate variables were expected to be involved in koa–grass interactions, we found no relationships. Grass suppression to a level that allows secondary succession was not occurring, indicating that planting monocultures is an ineffective tool to incite passive restoration. Rather, if diverse forest communities are desired, planting additional tree, shrub, and fern species could be used to amplify mechanisms that control invasive grasses and create conditions favorable for further forest species recruitment.

Acknowledgments

We thank Allie Trudgeon, Kim Fuhr, Jeff Stallman, and Gabe Runte for help in the field and lab; Christopher Warneke for plant identification and constant insight; Kevin Brinck for statistical advice; David Rothstein and the Institute for Pacific Island Forestry for providing lab space and access to equipment; Hawai'i Volcanoes National Park, The Nature Conservancy, Hawai'i Experimental Tropical Forest, USDA Forest Service, Hawai'i Department of Land and Natural Resources—Division of Forestry and Wildlife, US Fish and Wildlife Service, particularly Sierra McDaniel, Mel Johansen, Shalan Crysdale, Elliot Parsons, and Tom Cady for facilitating access to field sites. This manuscript was improved by comments from Lars Brudvig, Carla D'Antonio, Rich Kobe, Mike Walters, the Brudvig lab group, and three anonymous reviewers. Our fieldwork took place in the ahupua'a of Honohina, the moku of Hilo, on the mokupuni of Hawai'i, which are the ancestral, traditional, and contemporary lands of the Hawaiian people. We are grateful to have had access to these sites and to have learned from the forest. This research was funded by USGS, USDA Forest Service, and an NSF Graduate Research Internship Program (GRIP) grant to C.M.E. Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government. The findings and conclusions in this publication should not be construed to represent any official USDA determination or policy. Data can be found at Esch and Yelenik (2025).

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Supporting Information

The following information may be found in the online version of this article:

Table S1. Substrate age and soil classification for each study site.
Table S2. Koa stem density and basal area in koa plots at each study site.
Table S3. List of model statements from SEM.
Table S4. Estimates from generalized linear models using site-level means of grass biomass or basal area.
Table S5. Estimates from generalized linear models with a gamma distribution using site-level means of decomposition rate or koa litter biomass.
Table S6. Estimates from a generalized linear model with a gamma distribution using plot-level measures of koa litter biomass as response to plot level measures of decomposition rate with site as a random effect.
Table S7. Parameter estimates for each path included in the final SEM.
Table S8. Estimates from a generalized linear model with a gamma distribution using decomposition as a response leaf C:N ratios with site as a random effect.
Table S9. Results of a PERMANOVA examining composition of the understory of koa plots based on percent cover using Bray–Curtis dissimilarities with site as a random effect, based on 999 permutations.
Table S10. Percent cover based on visual estimation of understory of koa plots by site.
Figure S1. Photos of koa plots of varying densities from (a) Kona Hema, (b) Kahuku, and (c) Hakalau.
Figure S2. Relationships between: grass biomass in control plots and temperature (a) and precipitation (b), koa basal area in koa plots and temperature (c) and precipitation (d) at 10 sites.
Figure S3. Relationships between: decomposition rates and temperature (a) and precipitation (b), koa litter biomass and temperature (c) and precipitation (d).
Figure S4. Relationship between decomposition rate and koa litter biomass at eight sites.
Figure S5. Relationship between C:N ratio of koa leaves and decomposition rate at eight sites.
Figure S6. Composition of understory cover by koa basal area averaged across all sites and plots.

Coordinating Editor: Cara Nelson

Received: 28 October, 2024; First decision: 3 February, 2025; Revised: 23 July, 2025; Accepted: 23 July, 2025